Saptaparno Chakraborty

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Skills

Languages:	C, C++, JavaScript, Java, Python
Frameworks/Libraries:	Node.js, Express, React, Tailwind CSS
Databases:	MySQL, SQLite, MongoDB, PostgreSQL
Tools/Platforms:	Git, Linux, Vim, CMake, Docker, AWS, Postman
Soft Skills:	Curiosity, Problem-Solving, Discipline, Grit

Projects

SeptCrawler | *C++, Node.js* | *Link* Apr' 26 - Present

- Designed a modular search engine for learning-focused retrieval, targeting 4+ content categories (documentation, references, tutorials, discussion forums) with a multi-stage ingestion and retrieval pipeline.
- Completed initial high-level and low-level system design, defining 8+ core components across C++ search engine and Node.js orchestration layers, including crawler, parser, indexing engine, ranking pipeline, and inverted index structures for efficient query processing.
- Defined system data flow and component interactions across 2 major pipelines (ingestion + query flow), focusing on extensible ranking strategies and intent-aware result organization, preparing the system for implementation.

WorkshopHub | *Node.js, Express, MongoDB, React, JWT, bcrypt* | *Link* Nov' 25 - Mar' 26

- Developed a full-stack workshop management platform supporting 3 user roles, to streamline 5+ end-to-end workflows of educational workshops, including registrations, attendance, materials, feedback, and certificates.
- Designed and implemented 20+ RESTful APIs with JWT authentication, bcrypt-based password security, and role-based access control (Admin, Instructor, Participant), along with 7 MongoDB schemas modeling the complete lifecycle.
- Engineered a scalable system with secure access control and modular architecture, gaining hands-on experience in backend design, API structuring, and full-stack integration.

Priorify | *C++, STL, SQLite, FTXUI, CMake* | *Link* Jul' 25 - Apr' 26

- Built a priority-based task manager supporting 6 core operations (add, update, delete, list, complete, clear) to efficiently organize tasks with deterministic ordering and fast local execution.
- Implemented core functionality in C++ using std::priority_queue with a custom comparator and a heap → vector → heap rebuild approach, structured across 4 core source modules (db, task_manager, tui, main) and transitioned from a manual Bash-based build pipeline to a modular CMake system with 3 external system dependencies (sqlite3, pthread, dl), and FetchContent-based dependency management for FTXUI.
- Strengthened understanding of data structures and persistence design through SQLite integration, handling persistent storage across sessions, and improved usability by evolving the interface from basic console I/O to a terminal UI using FTXUI.

Training

CodeQuest: DSA Summer Bootcamp Jun' 25 - Jul' 25

Data Structures and Algorithms

- Focused on core and advanced Data Structures & Algorithms.
- Emphasized algorithmic optimization, problem-solving, and real-world software applications.
- Included daily hands-on coding, debugging, and complexity analysis sessions.

Certificates

- Computational Theory: Language Principle & Finite Automata Theory | *Infosys Springboard* Aug' 25
- Privacy and Security in Online Social Media | *NPTEL* Apr' 25
- Fundamentals of Network Communication | *University of Colorado* Nov' 24
- Introduction to Hardware and Operating Systems | *IBM* Sep' 24
- The Bits and Bytes of Computer Networking | *Google* Sep' 24
- Responsive Web Design | *freeCodeCamp* Nov' 23

Education

Lovely Professional University Punjab, India
Bachelor of Technology - Computer Science and Engineering; ***CGPA: 7.87*** *Aug' 23 - Present*

Railway H.S. School West Bengal, India
Intermediate (PCM); ***Percentage: 78*** *Apr' 22 - Mar' 23*

Railway H.S. School West Bengal, India
Matriculation; ***Percentage: 89*** *Apr' 20 - Mar' 21*